

Talagrand's Convexity and Subgaussian-Vector Problems Explicitly Answered by arXiv:2605.10908

Literature-status packet for arXiv:2602.22342

22 July 2026

Status: literature_already_answered. This packet records an explicit answer in a separate later paper. It does not claim a new proof produced by the run.

Source and exact questions

The source is Antoine Song, *Sum of Gaussian vectors and large sets*, arXiv:2602.22342v2 [1]. Problem 0.1 on arXiv PDF page 1 asks whether there is a universal integer q such that, for every n and every closed $A \subset \mathbb{R}^n$ with $\gamma_n(A) \geq 2/3$, there is a convex body K with

$$\gamma_n(K) \geq \frac{1}{2}, \quad K \subset \underbrace{A + \cdots + A}_{q \text{ times}}.$$

Problem 0.2 on the same page asks whether there is a universal integer $q' > 0$ such that every centered 1-subgaussian random vector $X \in \mathbb{R}^n$ can be represented as

$$X = G_1 + \cdots + G_{q'},$$

where each G_j is a standard Gaussian vector (the G_j need not be independent). Theorem 0.3 on page 2 states that these problems are equivalent.

The current source PDF already contains an “Added May 2026” note on page 2 identifying a subsequent paper as a solution of both problems.

Explicit answering paper

The separate paper Dongming (Merrick) Hua, Antoine Song, and Stefan Tudose, *On Talagrand's Convexity Conjecture*, arXiv:2605.10908v2 [2], restates both problems, cites [1], and says explicitly that it resolves them affirmatively.

Its Theorem 1.1 on arXiv PDF page 2 proves the stronger structural statement: if X is dominated in convex order by a standard Gaussian vector G , then there are three standard Gaussian vectors G_1, G_2, G_3 such that

$$X = G_1 + G_2 + G_3.$$

Corollary 1.2 combines this with a universal convex-order comparison for centered 1-subgaussian vectors. The paragraph immediately following the corollary explains how to remove the universal scaling by using a universal finite number of Gaussian summands. This is exactly an affirmative answer to Problem 0.2 of [1]. By the equivalence in Theorem 0.3 of [1], the convexity problem is also answered affirmatively; the answering paper also states a quantitative geometric version as Corollary 1.3.

Classification and scope

The identification is author-explicit, not an implication inferred only by this run: the answering paper includes the source author, cites the source, repeats the two questions, and states that they are resolved. It therefore belongs in `solutions/literature_already_answered/`, not in an original-solution bucket.

The full existential questions extracted from arXiv:2602.22342 are settled. The supporting paper separately notes that a constructive proof of its quantitative convex-body corollary remains open; that refinement is not part of the extracted questions classified here.

Evidence files

The packet includes the source PDF as `source_paper.pdf`. It includes the answering PDF as `supporting_paper_2605.10908.pdf`. The decisive locations are source pages 1–2 and supporting pages 1–2.

References

- [1] A. Song, *Sum of Gaussian vectors and large sets*, arXiv:2602.22342v2, 2026.
- [2] D. M. Hua, A. Song, and S. Tudose, *On Talagrand’s Convexity Conjecture*, arXiv:2605.10908v2, 2026.